

MBEYA UNIVERSITY OF SCIENCE AND TECHNOLOGY



FIELD PRACTICAL TRAINING AT TARURA ILALA TECHNICAL REPORT

STUDENTS NAME: .

REGISTRATION NUMBER:

DEPARTMENT:

LEVEL:

ACADEMIC YEAR:

INDUSTRY/FIRM:

UNIVERSITY SUPERVISOR:

Contents

ACKNOWLEDGMENT.....	i
SUMMARY	ii
LIST OF ABBREVIATION	iii
CHAPTER ONE	6
1.0. INTRODUCTION	6
1.1. BACKGROUND OF THE FIRM.....	6
1.2. ORGANIZATION SERVICES	6
1.3. ORGANIZATION STRUCTURE OF TARURA	7
1.4. VISION, MISSION AND CORE VALUES.....	7
1.5. OBJECTIVE OF THE FIELD ATTACHMENT.....	8
1.6. STRENGTH AND WEAKNESS OF THE FIELD ATTACHMENT PROGRAM.	8
CHAPTER TWO	10
2.0. ACTIVITIES PERFORMED.....	10
2.1. PLACEMENT AND WORKING ENVIRONMENT OF THE ORGANIZATION	10
2.2. GENERAL TASKS AND MAJOR TASKS PERFORMED.....	10
2.2.1. SOFTWARE INSTALLATION	11
2.2.2. LAN INSTALLATION	15
2.2.3. COMPUTER HARDWARE MAINTAINANCE.....	16
2.2.4. ANTIVIRUS INSTALLATION ACTIVATION AND UPDATING	17
2.2.5. WINDOW INSTALLATION	18
2.2.6. COMPUTER TROUBLESHOOTING	21
2.3. PROBLEMS FACED AND SOLUTIONS.....	22
2.4. EXPECTATIONS BEFORE GOING TO THE TRAINING	24
2.5. THINGS I LEARNT FROM TARURA	24
2.6. GAPS IN TEACHING BETWEEN THEORY AND PRACTICES	25
CHAPTER THREE	26
3.0. RECOMMENDATION	26
3.1. CONCLUSION.....	26
3.2. REFERENCES	27

ACKNOWLEDGMENT.

I thank GOD and my family for supporting me during all the period of practical training and all people who enabled the completion of my training.

I would like to give thanks to the management of TARURA for allowing me to conduct my field practice at their organization thus not only help me to increase my knowledge in ICT field but also help me to gain experience on how the real working environment is and it has really been a rewarding experience.

Specifically, I congratulate and thanks my field instructor **MR. Labibu Sultan** and **MS. Lilian** all **other staff members** for the better supervision on different activities during the practical training and thanks to the supervisor **MR Yamakili. P** for better instruction assistance in the preparation of the field report and teaching on how to behave on working place. Finally, great thanks should go to my fellow students by being hand in hand with me during field activities and enables me to learn new things and realize the strength of cooperation in life.

SUMMARY

The contents in this report it has highlighted what has been observed and done during the whole fieldwork as it covers the period from 16 August to 10 October 2021 at TARURA ILALA DSM. This field report contains the declaration section, the approval section, acknowledgement section, list of figure, list of abbreviation section and divided to three chapters. Chapter 1 based on the introduction in which it contains briefs explanation, background information of the organization, organization services, organization structure, vision, mission and core values and the objective of the field attachment. Chapter 2 contain the activities performed, chapter 3 conclusion and recommendation and finally is the reference.

LIST OF ABBREVIATION

PC-Personal Computer.

WI-FI-Wireless Federation.

OS-Operating System.

MS-Microsoft.

RJ-Registered Jack.

CMOS-Complementary Metal Oxide Semiconductor.

ROM-Read Only Memory.

CMD-Command Prompt

USB-Universal Serial Bus

RAM-Random access memory

CPU-central processing unit

ICT-Information and communication technology.

CD-Compact Disk

CHAPTER ONE

1.0.INTRODUCTION

1.1.BACKGROUND OF THE FIRM

The Rural and Urban Roads Agency (TARURA) was established in accordance with the Law on Government Agencies Chapter No. 245 and officially announced in the Government Gazette no. 211 of May 12, 2017.

The Agency started working officially on July 1, 2017 and now the Agency has reached approximately 5 years since its establishment. The main objective of the establishment of TARURA is to deal with the challenges that arose when the responsibility of the District Road Network Management was under the Local Government Authorities (MSM) 184, but also to deal with changes in the road sector with the intention of improving services and increasing efficiency.

The main responsibility of TARURA is to manage the maintenance, repair and construction of the District Road Network with a total of 144,429.77 Kilometers that were managed by 184 Local Government Authorities and coordinated by the President's Office TAMISEMI under the Department of Infrastructure. Other responsibilities of the Agency are as specified in the Agency Establishment Document No. 211 of May 2017, including measuring and conducting research on construction materials through the Central Equipment and Research Laboratory Unit, controlling the weight of vehicles on the road and protecting road reserve areas.

In addition, the Agency carries out its duties by entering into special agreements known as (Annual Performance Agreement 'APA') between it and the Road Fund for road maintenance works and the President's Office - TAMISEMI for development project work.

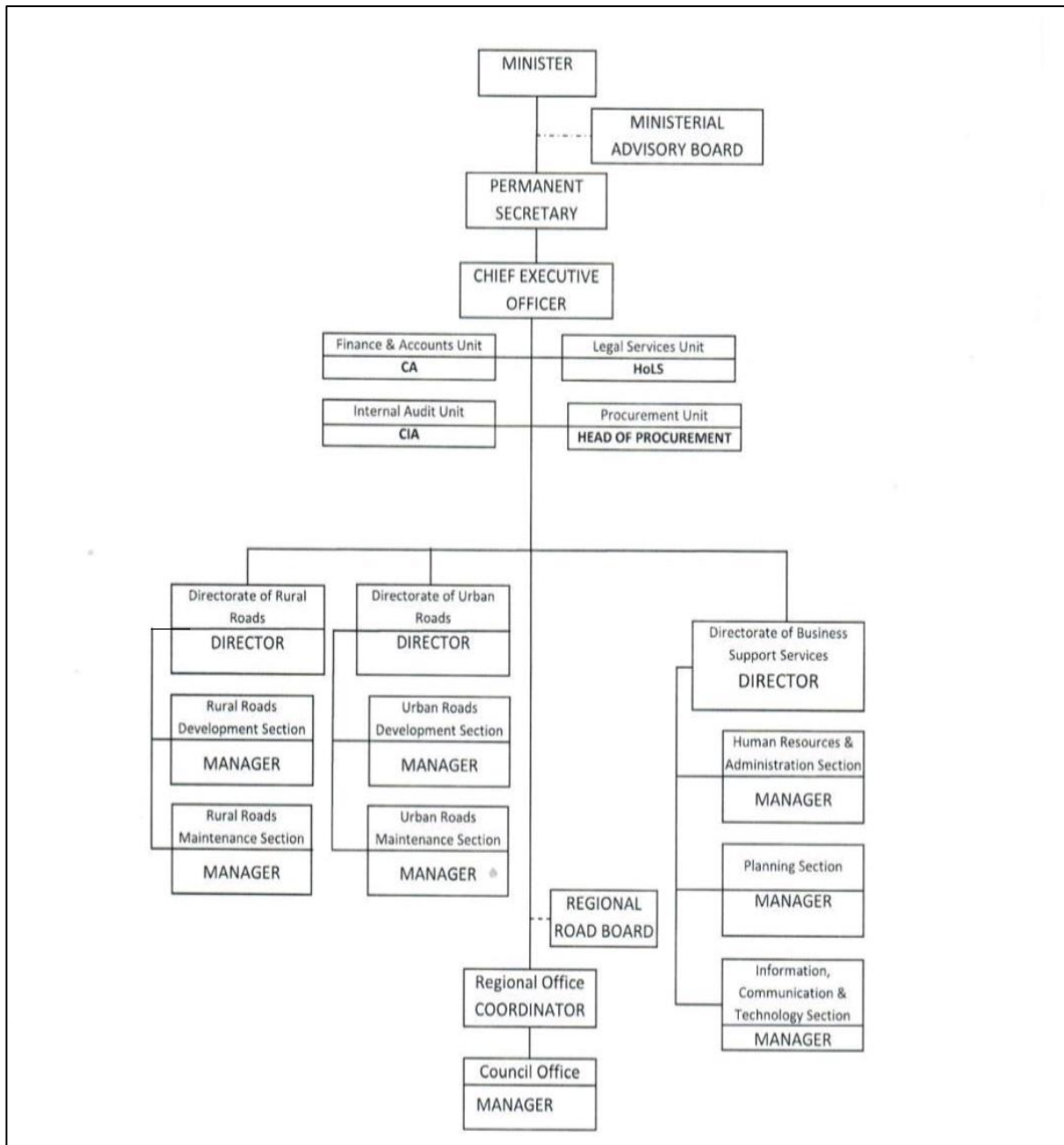
1.2.ORGANIZATION SERVICES

TARURA an organization that provides services to local and all areas in TANZANIA both rural and urban areas.

The services provided by the organization are:

- Road construction
- Professional advice
- Use laboratory services
- Road use

1.3.ORGANIZATION STRUCTURE OF TARURA



1.4.VISION, MISSION AND CORE VALUES

Vision

To be a leading Institution in managing the District's road network.

Mission

Planning, Designing, Building and maintaining the District's road network for sustainable social and economic results.

1.5.OBJECTIVE OF THE FIELD ATTACHMENT

The objective of the fieldwork is to put the theories into practical to gain the experience and clear knowledge of the different things concern a certain career. Also field attachment is the field based practical training experience that prepare trainees for the task or activities they are expected to perform on the completion of the training. Therefore, the following are the specific objective of the field attachment program.

Field attachment aimed at measuring the level of understanding by applying the theory concept and knowledge into practical work. This is due to that during the field training or fieldwork student will be able to put into practical the different theories that he or she learn into class.

Field attachment program also aim in increasing confidence to a student during the work. This is due to that since a student are assigned to perform the different activities by the onsite supervisor during the practical training this normally help a student to build the strong confidence at work environment also it assist the future development once he or she get the job.

Also field attachment aim to create the competence and develop a student to understand the different opportunity responsibilities and knowledge of the different employers also work ethics and on how the worker they behave in working area.

1.6.STRENGTH AND WEAKNESS OF THE FIELD ATTACHMENT PROGRAM. STRENGTH

The major strength of the field attachment program is that, the field attachment program normally its enable us to learn more in practical work also during my training I was gained a lot of skills and knowledge on how the activities are performed especially in information and communication technology department. For example on how to troubleshooting and maintenance of computer.

WEAKNESS

The major weakness of the field attachment program particular at the TARURA are few number of office facilities such as printer, photocopier and computer that can simplify work to be easily and completed at the short period of timer rather than using manual system.

So apart of the major strength and weakness of the field attachment program, the followings are the benefit derived from the field attachment program:

- Gain an experience of work and receiving the new Idea through the daily activities, which was assigned to me especially in information and communication technology.
- Sharing the Idea with workers in various matter concern the ICT. Also the program help in built the close link with the employee in order to have the better opportunity to the possible future employment.

CHAPTER TWO

2.0.ACTIVITIES PERFORMED

The chapter introduces the conceptual activities performed, it is generally indicates my placement in the organization, working environment at the organization during my field work, the duties I performed, challenges during my field work period, problem faced in executing the task and how I solved them.

2.1.PLACEMENT AND WORKING ENVIRONMENT OF THE ORGANIZATION

There was supervision on executing tasks for after being assigned to do a task and being given a brief introduction on how to do it. I was left to do it on my own or sometimes as a team. This kind of independence encouraged me and made me to have a self-trust on my own and confidence in performing the work and only asked for guidance when necessary.

In addition, there was a spirit of teamwork at TARURA, which entails that works are done in a cooperative manner to enhance efficiency, effectiveness and fast performance of works. In addition, the workers at TARURA were very friendly like to support each other in duties to make sure that the duties are completed on time.

2.2.GENERAL TASKS AND MAJOR TASKS PERFORMED

During my practical training at TARURA there were many tasks and duties assigned to perform. There were major tasks, critical tasks and urgently tasks performed by the supervisors.

The overall activities performed during my practical training were:

1. Software installation.
2. LAN installation.
3. Computer (PC, printer, keyboard and desktop) maintenance and repair.
4. Window installation.
5. Installation of Antivirus, Activation of window and window updating
6. Computer troubleshooting problems.
7. Opening and partitioning the hard disk.

2.2.1. SOFTWARE INSTALLATION

I was assigned to install a software application.

There are different types of software installations that include window installer installation, web based installation and single exe software installation. In addition, the installation (setup) of a computer program (including devices drivers and plugins) is the act of making the program ready or execution. Installation refers to the particular configuration of software or hardware with a view to make it usable with the computer

The following are the several steps in software installation

1. Preparation of the tools to be used

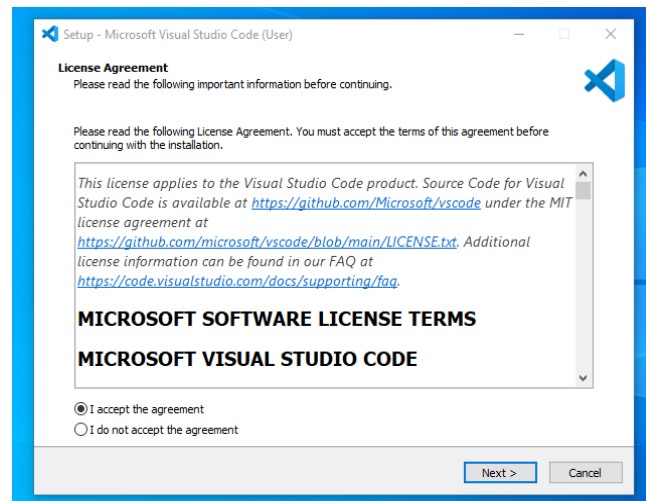
Ensure your computer meets the minimum system requirements for the software. Check if there any specific prerequisites or dependencies needed the software to run. Acquire the installation package by either downloading the software setup from the reliable source such as website or a trusted distributor. If use have a physical copy e.g. CD or flash.

2. Run the installer

Double click on the downloaded installation file or launch the installer from the physical media. Then you may see a prompt asking, “do you want to allow this app to make changes to your device?” selecting **yes** will run the setup file and begin installation on your computer.

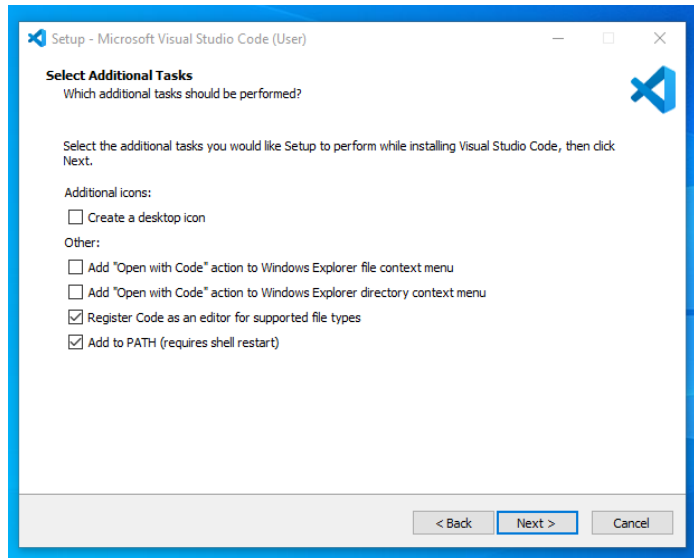
3. Accept license agreement

After the installer opens, it will ask you for accepting the terms and condition of the software to be installed. After reading the agreement click on, **I accept the agreement** and then click **next** button.



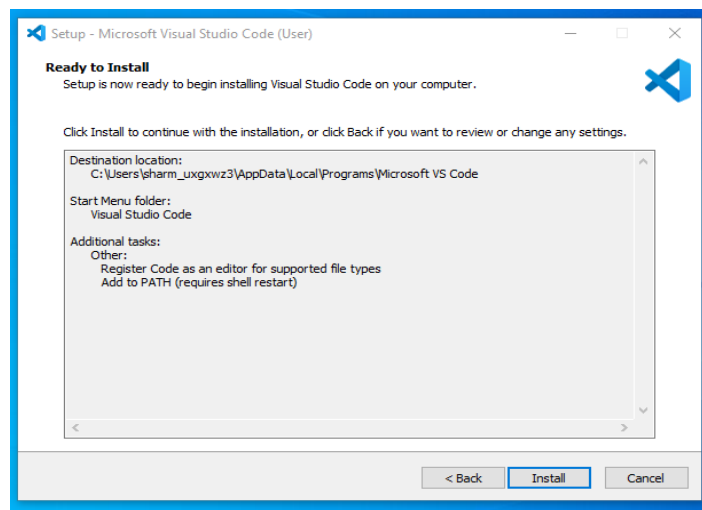
4. Choose the installing options

Some software installers offer customization options, such as choosing the installation location, components to install and additional features. Make selection based on your preferences and needs.



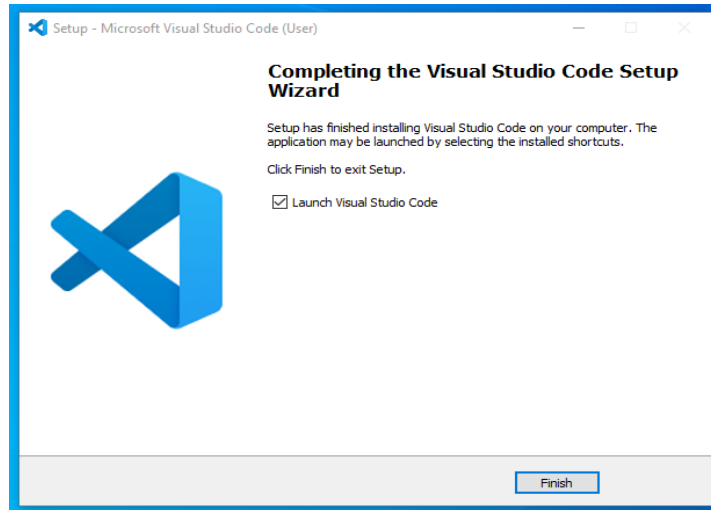
5. Start installation

Click the install button to begin installation process. The installer will copy necessary files to the specified location and it may take some time.



6. Finish installation

Once the installation is complete you will see a confirmation message some installers might prompt you to restart your computer before using the software.



7. Testing and verification

Launch the software to ensure it runs correctly. Test its various features and functions to make sure everything is working properly

NEW THINGS THAT I LEARN IN SOFTWARE INSTALLATION

Software installation thing enables the user to perform a certain task example Microsoft office, visual studio code and virtual DJ. So some of the installation may be successfully because of the following

- Available disk space
- Specific type of window required
- System requirements such as CPU, RAM, Graphics and other hardware specifications
- Download source from a reputable and official source to avoid malware
- Disable Antivirus/firewall to prevent interference with the installation
- Available internet connection

2.2.2. LAN INSTALLATION

Local Area Network (LAN) involves setting up a network infrastructure that allows computers and devices to communicate and share resources within a limited geographical area such as home, office or campus

The process includes connecting devices through cables or wireless technology, configuring network setting and ensuring secure and reliable communication

LAN installation are essential for enabling collaboration, resource sharing and data transfer among devices within a confined space



Colored Ethernet cable.

- ❖ Connecting an Ethernet cables from computer to computer devices.

NEW THINGS THAT I LEARN IN NETWORKING

Network is a system in which multiple devices are connected to each other to share information and resources. The following are the list of hardware's required to set up a network, which are Network Cables, Distributors, Routers, Internal Network Cards, External Network Cards and Universal Serial Bus (USB).

- ❖ Using cable tester to test each pin of the connector.
- ❖ Using the crimping tool.

Understanding the colored wires of an Ethernet cable, which are white/orange, orange, white/green, blue, white/blue, green, and white/brown, brown.

2.2.3. COMPUTER HARDWARE MAINTAINANCE

Computer maintenance refers to the regular tasks and practices undertaken to ensure the optimal performance, reliability and longevity of the computer systems

It involves proactive measures to prevent hardware and software issues and to address any existing problems

Printer maintenance

Printing all written documents and stamping them and arrange them to the file by sequential order to give good clarification of the documents in a specified date, this will help to ensure good environment to the organization. Since printers have many moving parts that can tear and wear out over time or through extended use, also moving parts can be affected by and other particles. In order to maintain good working condition of printer I repairing printers regularly to ensure the functionality of the printer in printing documents. In addition, that enables to avoid Loss of productivity and high repair



HP Laser Jet Printer

2.2.4. ANTIVIRUS INSTALLATION ACTIVATION AND UPDATING

Installation of symmetric point protection antivirus. During installation of this kind of antivirus, there are instructions to follow when installing this antivirus, which are -

1. First, you need to turn on your computer, if it is off.
2. Secondly uninstall the previous antivirus if present in your computer by click start and then select control panel, the control panel window will appear click programs and then select uninstall program or remove the program asymmetric end point protection.
3. Thirdly, after finish to uninstall the antivirus then restart your computer so that the program will be removed completely.
4. Then insert the cd of asymmetric end protection in your computer cd drive and the computer will automatically detect it and start read the files on the CD.
5. Select the setup for asymmetric end protection.
6. Then the license of agreement will appear click accept and then it will start installing the antivirus in your computer automatically.
7. Then click finish to complete installation.
8. Update the antivirus from the internet in order to protect your computer.
9. Click finish and then restart your computer to successfully install your antivirus.

The antivirus was to be installed in all new operating systems, the organization provided KASPESKY and SMADAV ant viruses, which required internet connection for its license activation. The first update took up to 10 minutes of update time, which used approximately 200MB. After the installation the antivirus was set for a schedule scan which would take place once a week while the computer is idle and custom scan whenever new mass storage device in connected to the system.

2.2.5. WINDOW INSTALLATION

One among the important things I learn during my field time was how to install the operating system especially Microsoft windows to computer devices as window 7 and window 10. There two ways of installing operating system either using cd or USB flash drive.

i. Installing OS using USB flash

If the operating system software came on a USB flash drive, you need to configure the computer to boot to a USB device as the first boot device. Once the computer is configured to boot to the proper device, the computer should load the operating system installation program and guide you through the install process. You will be asked questions along the way for configuration of basic settings, like date and time, user account name, and if you want to enable automatic operating, system updates. Go through the installation steps, answering questions and selecting the preferred options.

ii. Installing OS using CD or DVD

To install the computer's operating system using a CD or DVD, you need to configure your computer to boot from the CD/DVD drive. You can change the boot sequence in your BIOS setup, and setting the CD/DVD drive to be the first boot device. Some computers may also allow you to access the boot sequence directly at computer start up, without entering the BIOS, by pressing a specific key on the keyboard. The key to press differs for each computer, but is often the Delete key or one of the function keys.

Installation of Operating System (OS) windows 7 in Dell computer

Step 1:

Setup. If the CD has been inserted and the BIOS is set correctly, you will see a message telling you to “Press any key to boot from CD...” Press a key on the keyboard and the setup process for Windows 7 will launch. Your system may automatically launch the setup program without asking you to press any key.

Step 2:

Watch the Windows files load. After completion, the Windows 7 logo will appear. No files have been altered on your computer yet. Your data will be deleted in later steps.

Step 3:

Choose your preferences. You will be prompted to confirm your Language, Time & Currency format, and Keyboard or input method. Select the appropriate options for you and click next.

19

Step 4:

Click Install Now. Do not click repair computer, even if you are repairing a computer by reinstalling Windows. Once clicked, Setup will begin loading the files it needs to continue.

Step 5:

Read and accept the terms. In order to advance, you must indicate that you have read and agree to Microsoft's terms of use. Be sure to read it so that you know your rights and limitations as a user.

Step 6:

Choose Custom install. This will allow you to perform a clean installation. Even if you are upgrading a previous version of Windows, it is highly recommended that you perform a clean install. Performing an Upgrade will often lead to devices and programs not operating efficiently and effectively.

Step 7:

Delete the partition. A window will open asking where you would like to install Windows. In order to perform a clean install, you need to delete the old partition and start with a clean slate. Click "Drive options (advanced)." This will give you the ability to delete and create partitions.

- Select the partition of your existing operating system and click the Delete button.
- If you are installing an operating system for the first time on this hard drive, then there will be no partitions to delete.
- If your hard drive has multiple partitions, be sure to delete the correct one. Any data on a deleted partition is lost for good.
- Confirm the deletion process.

Step 8:

Select the Unallocated Space and click next. There is no need to create a partition before you install Windows 7, this is done automatically.

Step 9:

Wait while Windows installs files. The percentage next to Expanding Windows files will steadily increase. This part of the process can take up to 30 minutes.

- Windows will automatically restart your computer when finished.
- Setup will launch again, and a message will tell you that Setup is updating registry settings.

- Setup will then configure your computer's services. This happens every time you start Windows, but will happen in the background next time.
- A window will open letting you know that Windows is completing the installation. Your computer will restart again when this is complete.

20

- Setup will now load drivers and check video settings. This part does not require any input from you.

Step 10:

Enter your user name and computer name. Your username will be used to log in to the computer and personalize your account. Your computer name is the name that your computer will display on the network.

- You can add more users later through the Windows 7 control panel.
- Windows will ask you for a password. This is optional but highly recommended, especially if the computer will be accessible by users other than yourself. If you'd rather not have a password, leave the fields blank and click Next

Step 11:

Enter your product key. This 25-character key came with your copy of Windows. Check the "Automatically activate Windows when I'm online" to have Windows automatically verify your key the next time it is connected to the internet.

Step 12:

Select your Windows Update option. To make sure that your copy of Windows runs securely and stably, it is highly recommended that you choose one of the first two options. The first option will install all updates automatically; the second option will prompt you when important updates are available.

Step 13:

Select your date and time. These should be correct already as they are linked to your BIOS, but you can change them now if they are not. Check the box if your area observes Daylight Savings.

Step 14:

Select network preferences. If your computer is connected to a network, you will be given an option to identify that network. Most users will select Home or Work network. If your computer is being used in a public place, select Public network. Mobile broadband users should always select Public network. Windows will now attempt to connect your computer to the network. This process is completely automated.

Step 15:

Explore your desktop. After one final loading screen, your new Windows 7 desktop will appear. Installation is now complete.

2.2.6. COMPUTER TROUBLESHOOTING

Computer troubleshooting is the process of identifying and resolving issues that arise in computer systems. It involves systematically analyzing symptoms isolating causes and implementing solutions.

This may involve checking hardware connections, updating software and drivers, scanning, reviewing error messages and performing tests to pinpoint problems.

The goal is to restore normal functionality and prevent future issues by addressing root causes Effective troubleshooting requires a logical approach, attention to detail and familiarity with computer components and software

Computer troubleshooting problem:

1. Identify the problem
2. Isolate the cause
3. Check connections
4. Restart or reboot
5. Update software and drivers
6. Scan for malware and viruses
7. Review error messages
8. Test in safe mode
9. Check system recourses
10. Restore from backup
11. Clear cache and temporary files
12. Check for overheating

THINGS I LEARNT FROM TROUBLESHOOTING

What I learned in troubleshooting is troubleshooting requires a systematic approach, patience and attention to detail where by following the logical steps you can easily and effectively identify and resolve a wide range of computer related issues.

2.3.PROBLEMS FACED AND SOLUTIONS

PROBLEMS FACED

- **Insufficient working facilities**

This is the one among problems which I faced during my practical training example inadequate computers and office space, and most of time there was no enough tools to accomplish, my duties

- **Poor networking systems**

In TARURA, organization there is poor networking system was common. The problem is that there are use of old equipment to run new programs that require more power and processing capacities than the old equipment can handle. Also there was the problem of internet occurred most of time which make some kind of tasks not to be carried on as expected this results in to delaying of completing works assigned.

- **The time set for practical**

The period of two month for practical training is not enough for a student to gain much knowledge and skills from the office compared to the theoretical knowledge taught in college.

- **Financial problem**

The fund allocated for the study was not enough compare to the needs and sometimes delayed this leads the student to be forced to do it under a very tight budget thus may resulted in boycotts eventually unnecessary confrontations.

SOLUTIONS TO THE PROBLEM FACED

- **Poor networking systems**

The organization should improve their network system by considering the following, first to change old equipment to new one that can run new programs that require more power and processing capacities than the old equipment can handle. To make sure that there is availability of internet at any time during time to work. In addition, the Organization must adapts the use of new technology so that to make their works to move smoothly and to be completed on appropriate time.

- **The budget should be increased**

In order for a student to do better on his/her practical training, the fund should be increased so that can satisfy the important requirement to the student during practical training.

2.4.EXPECTATIONS BEFORE GOING TO THE TRAINING

- By my side I expected that most of tasks and duties would be difficult to handle contrary to my expectation's things were not tough as I thought because most of duties and tasks were just normal and the ones that I couldn't manage I was clearly instructed.
- The other thing that I was expected is this, that different officers including my internal supervisor within the organization will train me in the way that their not friendly with me that can make me to be afraid them, but I thanks God things was going against my expectations and the officers including my supervisors was very friendly to me which made me to have more confidence in the work and also to learn more things by sharing my views, ideas, contributions and also to ask many different questions concerning my training.
- I was also expecting to acquire new knowledge and practical experience, also improving my confidence in solving problems. In addition, this was successfully as I expected because I learned many different new things.
- Unfortunately also some of my expectation was some met and other were not, in this way I was to cope with the situation and bided by the rules and regulations for the performance of works assigned.

2.5.THINGS I LEARNT FROM TARURA

- In TARURA, I learnt many new things such as what are being taught in class are very important and helpful even at work places, because most of the things are applicable even at work.
- In addition, I learnt to be punctual. TARURA is a very busy organization and work has to be done early as much as possible especially ICT support section, to arrive early in the morning at 07:00am so as to ensure that no network problems and many other including helping other officers whenever they have problems.
- I was also learnt how to work efficiently under minimal supervision since I was deployed at different offices where there was minimal supervision but worked fast and efficiently.

25

- Also I was learnt about how to behave when am at work with my fellow workers that we should respect each other, and also to follow the rules and the regulations of the organization, thus to work properly without violating those rules and regulations, also to make sure that things of the company remain inside the organization.

2.6.GAPS IN TEACHING BETWEEN THEORY AND PRACTICES

In teaching between theory and practice, there is a gap as I explain here below:

- A theory is a complete and relational type of abstract or generalizing thinking, or the result of that thinking depend on context the result might, for example including generalized explanation of the nature of work. The theory which I learn in class is the same but not similar in practical during my Feld training.
- This is because in practical I have gained more knowledge and experience and confidence, knowing how different instruments by viewing and touching them example, blower, crimping tool, RJ-45, flat screwdrivers, switch, patch panel, Ethernet cables, and trunks.
- It is very simple and easier to understand through practical than theory because it took me a little time to understand how different works in Information Technology (IT) can be done

at TARURA, because I was taught in practical way. Overall, theory and practical need each other in accomplishing tasks and major understanding.

CHAPTER THREE

3.0.RECOMMENDATION TO THE FIRM

The organization at TARURA should make daily maintenance of their computer devices such as desktops and printer, because the devices sometimes failure to operate better even corrupt due to lack of long time maintenance like blowing dust and security update.

The organization at TARURA should control their system accordingly because sometimes the network of the system seems to be low and some networking device in the organization failure to get the internet.

TO THE INSTITUTE

The institute Mbeya University of Science and Technology (MUST) that they should search field for training to the student basic on difference course they undertake cause student they face and gate a hard time during searching for field in different company or organization

3.1.CONCLUSION

I can conclude by saying that I enjoyed the training since I got new knowledge, skills about computer technology as well as on how to interact with other people at the organization of different departments.

Through practical learning during the field training/practical many of the activity makes me aware on the application of the theoretical knowledge, which also gives me experience and exposure to my career.

The thing, which makes me enjoy, was the instruction given from the supervisor on the practical area on how to perform different activities procedural, and cooperation from my fellow students, which brings up work motivation.

3.2.REFERENCES

Keewatin, C. (2003), *Basic Computer Troubleshooting and Maintenance Guide*, Career development Corporation press. New York.

J, A. (2012). *A+ Guide to Managing and Maintain your pc*. US: United State of America.

Richard(2015,June20).Wikipedia.RetrievedfromAvailableh20564.www2.hp.com/hpsc/swd/public/readIndex?sp4ts: <http://www.wikipedia.com>

Richard, D. M. (2003). *Data Communications and Computer networks for Scientists and Engineers*. New Delhi: Prentice-Hall of India private ltd.

Slon, J. D. (2001). *Network Troubleshooting Tools*. Kenya: O'reilly.

Stifiel, B. J. (2005). *Advanced Configuration and Troubleshooting*. Syngress. New Delhi: Prentice-Hall of India private ltd.

